Singular events, such as the births and deaths of individuals can make it difficult to directly perceive any probabilities that are hidden in a plethora of other circumstances. George R. Price[1] modelled selection as a truly mechanistic process. In particular his covariance formulation of the mathematical theory of Natural Selection is regarded as conceptually elucidating (Price, 1970). I will derive Price's equation, and in parallel use Popper's theorems to derive a dynamical system which make it possible to calculate the change of gene frequencies over time.

We assume a single locus with two alleles, A and a, and calculate the change in frequency of the gene A from one generation to the next, ΔQ. In this case his equation looks like this:

$$\Delta Q = \frac{\operatorname{cov}(z, q)}{\overline{z}} \tag{1}$$

Here z_i is the number of successful gametes from individual i ($i = 1 \dots N$ where N is the population size) and q_i is the frequency of A in that individual, $q_i = 0, 1/2, 1$. $\overline{z}$ and $\overline{q}$ are the arithmetic means of the number of successful gametes and frequencies of A over all the N *individuals* in the population, respectively. Note that Price adds an alternative formulation

$$\Delta Q = \frac{\beta_{zq}\sigma_q^2}{\overline{z}} \tag{1a}$$

Here β_{zq} is the slope of a regression of z_i, the succesful games on q_i, the frequency of gen A in individual i. Here we can see how natural selection operates using standard statistical analysis. I give a hypothetical example in Figure 5.

Price considers two populations P_1 and P_2, where the former represents the parents and the latter their offspring. We first consider the P_1. Let g_i be the dose of the gene A in individual i. $g = 0, 1, 2$ for a diploid species. Further, let n_z the zygotic ploidy of the species. For a diploid species $n_z = 2$. The frequency of the gene A in P_1 is

$$Q_1 = \frac{\sum_{i=1}^{i=N} g_i}{n_z N} = \frac{\sum_{i=1}^{i=N} n_z q_i}{n_z N} = \overline{q} \tag{2}$$

where we take the sums over all members of population P_1.

In traditional population genetics you do this differently. Here you assume (i) Mendel's law of inheritance and (ii) random mating. (Price assumes nothing of this kind when deriving his equation.) Imagine that we pick up each individual and take a note of its genotype, if it is *AA*, *Aa* or *aa* (basically this is what Price does). From those data we can easily calculate the gene frequency. Now we use the traditional assumptions and Equation (2) to calculate $\overline{q}$. We knew the frequency Q_1 of A in the population P_1. Let us now form the population P_1 as zygotes from the gametes from their parents. This is an urn experiment. We draw two A with probability Q_1^2, it follows from the multiplication theorem, Equation (P3). Likewise, we pick up one A followed by a a with the probability $Q_1(1-Q_1)$ and one a followed by a A with probability $(1-Q_1)Q_1$—both using the multiplication theorem Equation (P3) and then we sum them together per Equation (P2). Since the two are identical, sum them together and get $2Q_1(1-Q_1)$.

$$Q_1 = \frac{2Q_1^2 N + 2Q_1(1-Q_1)N}{2N} = \overline{q} \tag{2'}$$

[1] Price was American population geneticist and before that he had a career as a physical chemist and as science journalist. He worked in the Manhattan project with the physical and chemical properties of Uranium 235, contracted thyroid cancer during the 1950s. He died by his own hand 1975, possibly induced by hypothyroidism which is both a physical and psychiatric condition following thyroid hormone deficiency. He lived among homeless in London and gave away all his possessions, while working on the evolution of altruism at the Galton Laboratory, University College, London (Harman, 2011).

Here we have used 2 for g_i for all those that are homozygous and 1 for those that are heterozygous. This way to go from gene frequencies to genotype ones is called Hardy-Weinberg's law:

$$\begin{aligned} AA&: \quad Q^2 \\ Aa&: \quad 2Q(1-Q) \\ aa&: \quad (1-Q)^2 \end{aligned}$$

There is nothing more than Popper's theorems (or the corresponding traditional ones), behind this exercise.

Now, turn to the frequency of A in the offspring population P_2 (the next generation). That frequency is determined by the number of successful gametes z_i from each individual i, but also the number of A genes g'_i. We also have to take into account the gametic ploidy n_G. For a diploid species the ploidy is $n_z = 2$ (see above) and consequently the gametic is $n_G = 1$. The frequency of A in individual gametes produced by individual i is then defined by

$$q'_i = \begin{cases} g'_i/z_i n_G & \text{if } z_i \neq 0 \\ q_i & \text{if } z_i = 0 \end{cases} \tag{3}$$

Let $\Delta q_i = q'_i - q_i$ for short. It is the difference between the frequency of A genes per successful gamete from individual i and the frequency of that gene in that individual. This quantity is very important, and I will return to its significance below.

Also, recall that the population covariance between two stochastic variables, lets call them z and q, are defined as

$$\text{cov(z, q)} = \frac{\sum_{i=1}^{N}(z_i - \overline{z})(q_i - \overline{q})}{N} = \frac{\sum_{i=1}^{N} z_i q_i}{N} - \overline{z}\,\overline{q} \tag{4}$$

Let Q_2 be the frequency of A in the offspring population P_2. Then

$$Q_2 = \frac{\sum_{i=1}^{N} g'_i}{\sum_{i=1}^{N} z_i n_G} = \frac{\sum_{i=1}^{N} z_i n_G q'_i}{\sum_{i=1}^{N} z_i n_G} = \frac{\sum_{i=1}^{N} z_i q'_i}{N\overline{z}} \tag{5a}$$

$$= \frac{\sum_{i=1}^{N} z_i q_i}{N\overline{z}} + \frac{\sum_{i=1}^{N} z_i \Delta q_i}{N\overline{z}} = \overline{z}\,\overline{q} + \frac{\text{cov}(z,q)}{\overline{z}} + \frac{\sum_{i=1}^{N} z_i \Delta q_i}{N\overline{z}} \tag{5b}$$

$$= \overline{q} + \frac{\text{cov}(z,q)}{\overline{z}} + \frac{\sum_{i=1}^{N} z_i \Delta q_i}{N\overline{z}} \tag{5c}$$

Now we are approaching the end of the derivation. Subtract Equation (2) from (5c)

$$\Delta Q = Q_2 - Q_1 = \frac{\text{cov}(z,q)}{\overline{z}} + \frac{\sum_{i=1}^{N} z_i \Delta q_i}{N\overline{z}} \tag{6}$$

All this was to derive Equation (1) from first principles. The second term in Equation (6) is not a bug but a feature. It is the mean value of the differences between the frequency of A genes per successful gamete from individual i and the frequency of that gene in that individual. This is a gross simplification; we assume that all individuals with the same genotype are exactly the same in all other aspects. We assume that all those other aspects somehow even out. The genotypes have become what von Mises refer to as *classes* or *collectives* (von Mises, 1957).

Another factor that will make it non-zero is genetic drift, also known as sampling error. There are also a range of other factors, like sexual selection (Fisher, 1930, p. 146–156). Selection can also operate through meiotic drive and segregation distortion, when a gene is trying to get an upper hand in the Battle of Transmission (Lindholm *et al.*, 2016) rather than in improving survival and reproduction in general. That term is an argument for the propensity interpretation of probability.

The traditional population genetics use another formulation. We start with a variation of Equation (4)

$$\mathrm{cov(z,q)} = \frac{\sum_{i=1}^{N} z_i q_i}{N} - \overline{z}\,\overline{q} = \frac{NQ_1^2 z_{AA} + N2Q_1(1-Q_1)z_{Aa}\,\frac{1}{2}}{N} - \overline{z}\,\overline{q} \tag{4'}$$

where we assume Hardy-Weinberg equilibrium and remember the definition of q_i above. Also we replace the index i on the z_i (the number of successful gametes from individual i) with their genotypes. I also replace Q_1 and $\overline{q}$ with the letter p (they are equal under the given assumptions)

$$\mathrm{cov(z,q)} = p^2 z_{AA} + p(1-p)z_{Aa} - \overline{z}\,\overline{q}$$

Substituting this into Equation (1) (using the p rather than Q) we get

$$\Delta p = \frac{p^2 z_{AA} + p(1-p)z_{Aa}}{\overline{z}} - p \tag{1'}$$

which is derived from the Price equation rather than the conventional way, as Okasha (2024) does it.

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